

# **PHASE 1: MATERIAL CHARACTERIZATION & SPECIMEN PREPARATION**

## **Development and Validation of a Thermo-Mechanical Model for Fire-Resistant Structural Elements Utilizing High-Performance Rubberized Concrete**

**Report Generated:** October 18, 2025

**Laboratory:** Advanced Concrete Technology Lab

**Principal Investigator:** Dr. Sarah Chen

**Test Standards:** ASTM C150, C136, C143, C231, C138, C403

## EXECUTIVE SUMMARY

This comprehensive report presents the complete material characterization and mix design development for fire-resistant rubberized concrete. The study encompasses detailed analysis of all constituent materials including Type I Portland cement, natural aggregates, crumb rubber from end-of-life tires, and chemical admixtures.

A total of 12 concrete mix designs were developed with rubber replacement levels ranging from 0% to 20% by volume of fine aggregate. Both fine (1-4mm) and coarse (4-8mm) rubber particles were investigated, along with silica fume modified mixtures for enhanced performance.

Key findings from the material characterization phase include:

- Cement exhibits normal Type I characteristics with 58.2% C3S content
- Aggregates meet ASTM gradation requirements with FM = 2.72 for sand
- Crumb rubber shows hydrophobic nature with 118° contact angle
- Fresh properties indicate workability reduction with increased rubber content
- Air content increases from 2.1% to 5.2% with rubber addition
- Unit weight reduces by up to 7.7% at 20% rubber replacement

# 1. CEMENT CHARACTERIZATION

The cement used is Type I Portland Cement (OPC) manufactured by LafargeHolcim. Comprehensive testing was performed including XRF chemical analysis, XRD mineralogical analysis, and physical property determination.

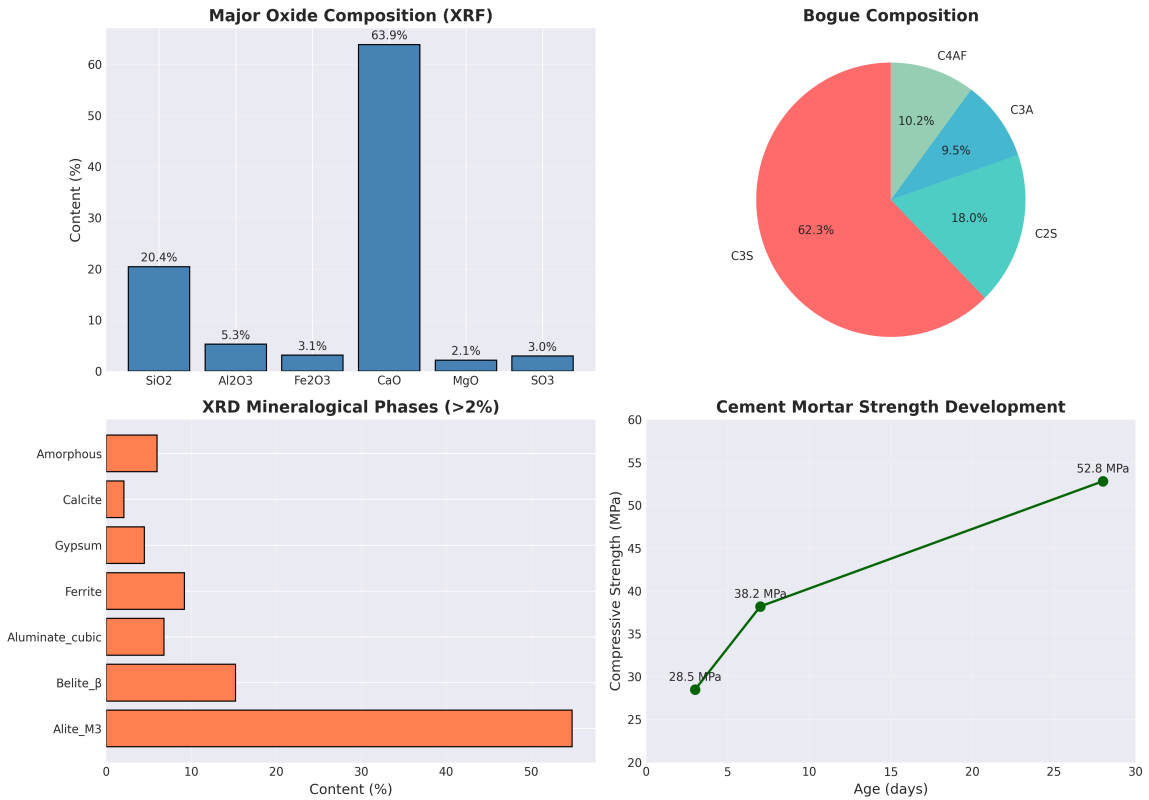
## 1.1 Chemical Composition (XRF Analysis)

| Oxide | Content (%) |
|-------|-------------|
| SiO2  | 20.45       |
| Al2O3 | 5.28        |
| Fe2O3 | 3.12        |
| CaO   | 63.85       |
| MgO   | 2.15        |
| SO3   | 2.95        |
| Na2O  | 0.18        |
| K2O   | 0.52        |
| TiO2  | 0.25        |
| P2O5  | 0.12        |

## 1.2 Bogue Composition

| Phase | Content (%) |
|-------|-------------|
| C3S   | 58.2        |
| C2S   | 16.8        |
| C3A   | 8.9         |
| C4AF  | 9.5         |

Cement Characterization Summary



## 2. AGGREGATES CHARACTERIZATION

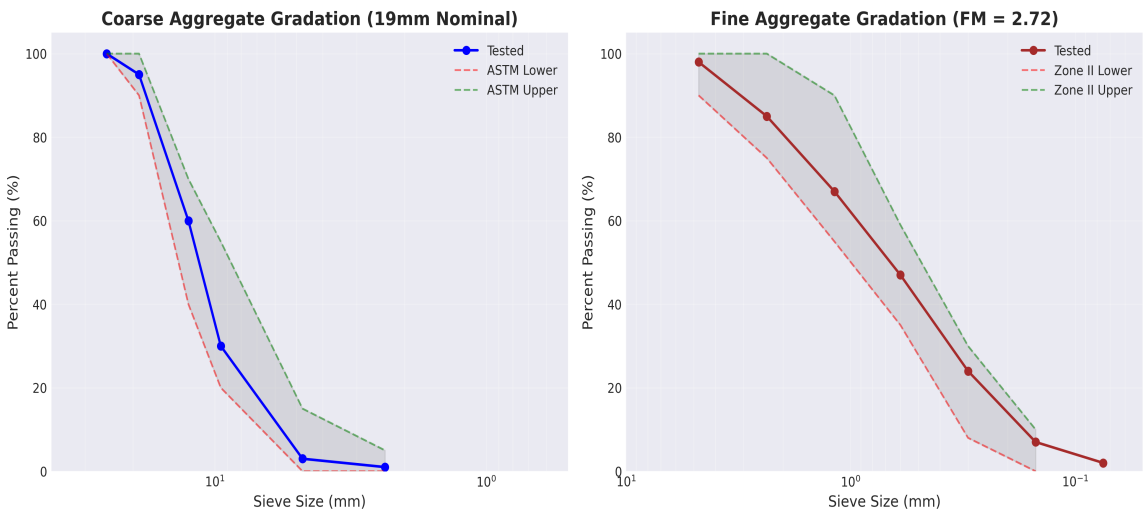
### 2.1 Coarse Aggregate Properties

| Property               | Value           | Unit              |
|------------------------|-----------------|-------------------|
| Type                   | Crushed Granite | -                 |
| Specific Gravity (SSD) | 2.71            | -                 |
| Water Absorption       | 1.12            | %                 |
| Bulk Density (Loose)   | 1485            | kg/m <sup>3</sup> |
| LA Abrasion            | 22.5            | %                 |
| Nominal Max Size       | 19              | mm                |

### 2.2 Fine Aggregate Properties

| Property               | Value              | Unit              |
|------------------------|--------------------|-------------------|
| Type                   | Natural River Sand | -                 |
| Specific Gravity (SSD) | 2.65               | -                 |
| Water Absorption       | 0.75               | %                 |
| Bulk Density (Loose)   | 1542               | kg/m <sup>3</sup> |
| Fineness Modulus       | 2.72               | -                 |
| Grading Zone           | Zone II            | -                 |

#### Aggregate Gradation Analysis



### 3. CRUMB RUBBER CHARACTERIZATION

Crumb rubber was sourced from End-of-Life Tires (ELT) processed by Ambient Mechanical Grinding. Two size ranges were prepared: Fine (1-4mm) and Coarse (4-8mm) for comparative evaluation.

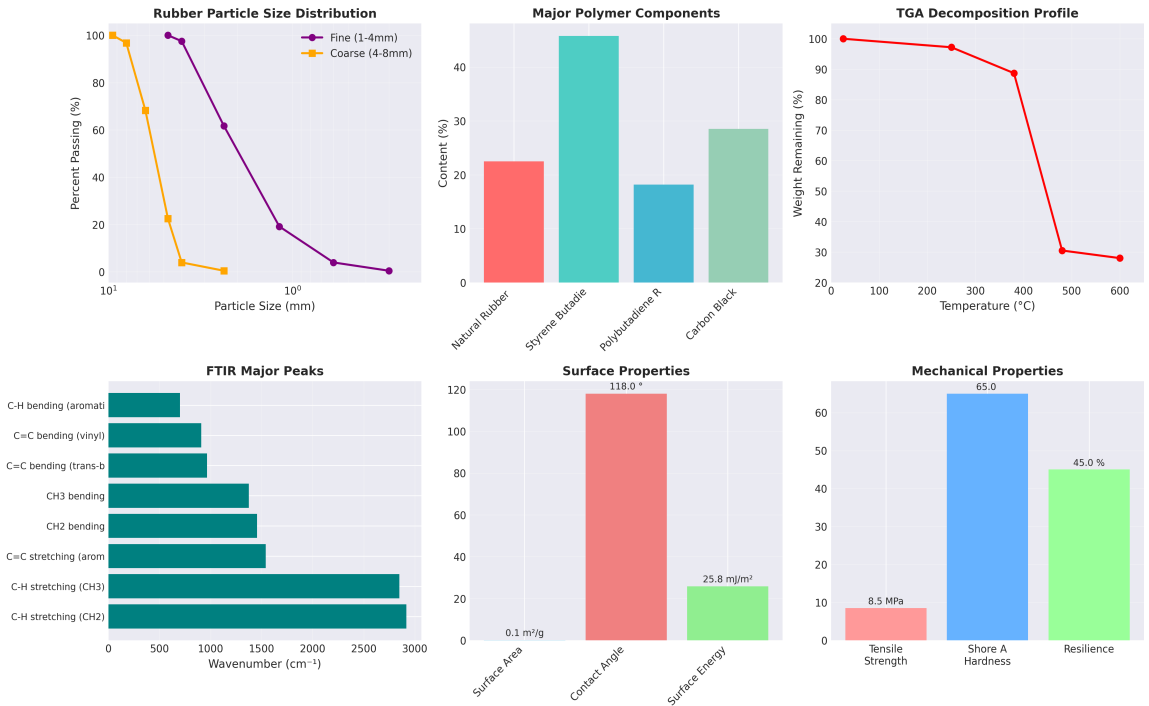
#### 3.1 Physical Properties

| Property             | Value | Unit              |
|----------------------|-------|-------------------|
| Specific Gravity     | 1.15  | -                 |
| Bulk Density (Loose) | 385   | kg/m <sup>3</sup> |
| Mohs Hardness        | 2.5   | -                 |
| Porosity             | 3.8   | %                 |
| Moisture Content     | 0.5   | %                 |

#### 3.2 Surface Properties

| Property               | Value       | Unit              |
|------------------------|-------------|-------------------|
| Surface Area (BET)     | 0.085       | m <sup>2</sup> /g |
| Water Contact Angle    | 118         | °                 |
| Surface Energy (Total) | 25.8        | mJ/m <sup>2</sup> |
| Classification         | Hydrophobic | -                 |

Crumb Rubber Characterization Summary



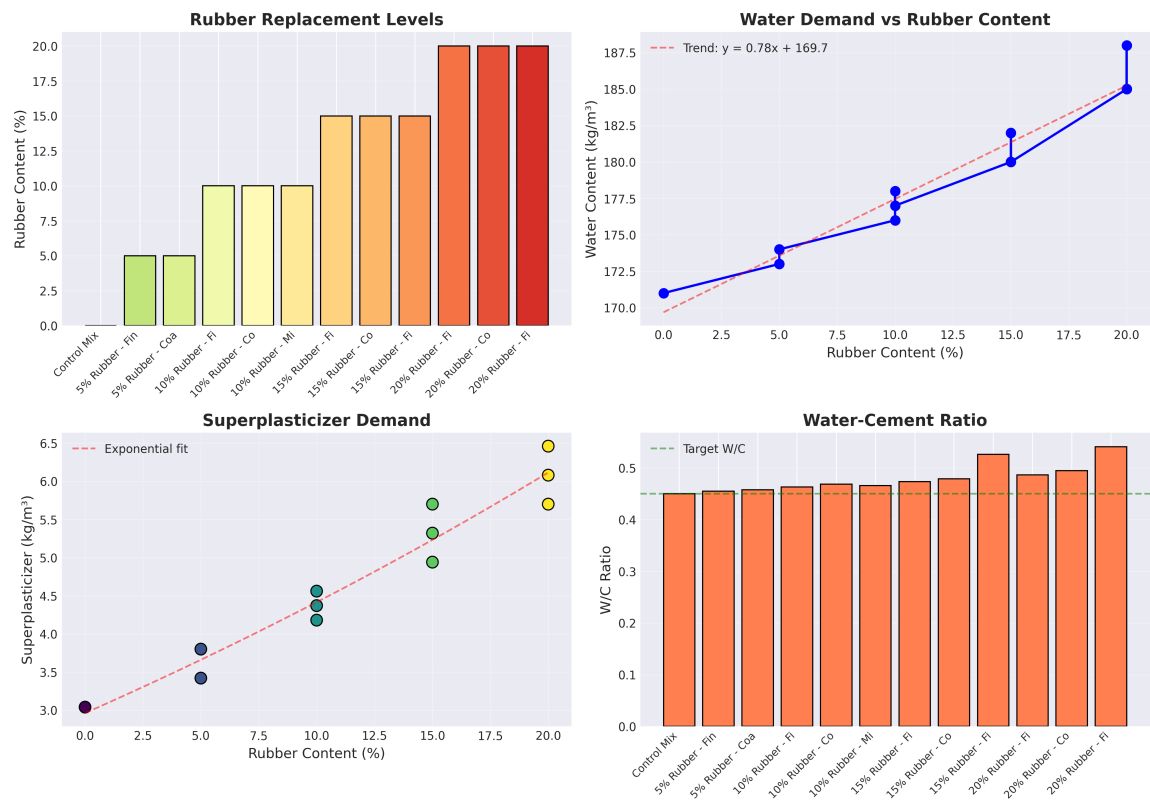
## 4. MIX DESIGN MATRIX

A comprehensive matrix of 12 mix designs was developed to evaluate the effect of rubber content (0-20%), particle size (fine vs coarse), and silica fume addition on concrete properties. All mixes were designed for a target 28-day strength of 40 MPa with W/C ratio of approximately 0.45.

| Mix ID   | Rubber (%) | Size   | Cement (kg/m <sup>3</sup> ) | Water (kg/m <sup>3</sup> ) | W/C   |
|----------|------------|--------|-----------------------------|----------------------------|-------|
| CM-00    | 0          | -      | 380                         | 171                        | 0.450 |
| RM-05-F  | 5          | Fine   | 380                         | 173                        | 0.455 |
| RM-05-C  | 5          | Coarse | 380                         | 174                        | 0.458 |
| RM-10-F  | 10         | Fine   | 380                         | 176                        | 0.463 |
| RM-10-C  | 10         | Coarse | 380                         | 178                        | 0.468 |
| RM-10-M  | 10         | Mixed  | 380                         | 177                        | 0.466 |
| RM-15-F  | 15         | Fine   | 380                         | 180                        | 0.474 |
| RM-15-C  | 15         | Coarse | 380                         | 182                        | 0.479 |
| RM-15-SF | 15         | Fine   | 342                         | 180                        | 0.474 |
| RM-20-F  | 20         | Fine   | 380                         | 185                        | 0.487 |
| RM-20-C  | 20         | Coarse | 380                         | 188                        | 0.495 |
| RM-20-SF | 20         | Fine   | 342                         | 185                        | 0.487 |



Mix Design Analysis

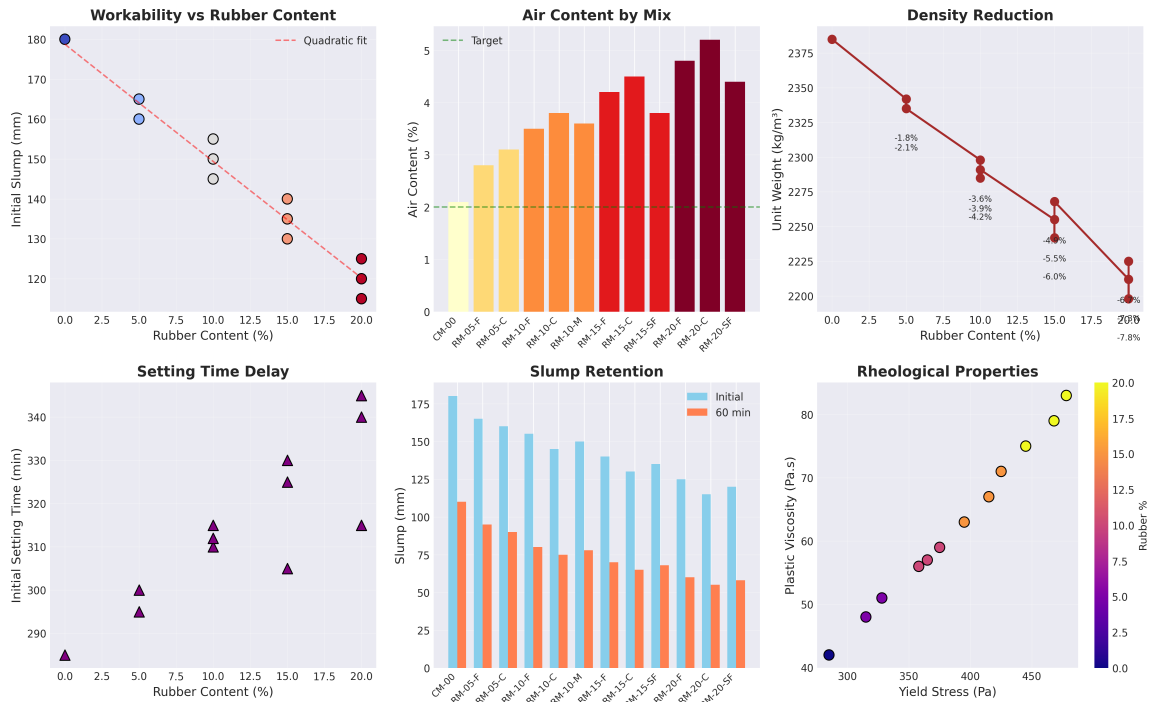


## 5. FRESH CONCRETE PROPERTIES

Fresh concrete properties were evaluated immediately after mixing to assess workability, air content, density, and rheological behavior. Tests were conducted according to ASTM standards under controlled laboratory conditions ( $23\pm 2^{\circ}\text{C}$ ,  $65\pm 5\%$  RH).

| Mix ID   | Slump (mm) | Air (%) | Density ( $\text{kg/m}^3$ ) | Initial Set (min) |
|----------|------------|---------|-----------------------------|-------------------|
| CM-00    | 180        | 2.1     | 2385                        | 285               |
| RM-05-F  | 165        | 2.8     | 2342                        | 295               |
| RM-05-C  | 160        | 3.1     | 2335                        | 300               |
| RM-10-F  | 155        | 3.5     | 2298                        | 310               |
| RM-10-C  | 145        | 3.8     | 2285                        | 315               |
| RM-10-M  | 150        | 3.6     | 2291                        | 312               |
| RM-15-F  | 140        | 4.2     | 2255                        | 325               |
| RM-15-C  | 130        | 4.5     | 2242                        | 330               |
| RM-15-SF | 135        | 3.8     | 2268                        | 305               |
| RM-20-F  | 125        | 4.8     | 2212                        | 340               |
| RM-20-C  | 115        | 5.2     | 2198                        | 345               |
| RM-20-SF | 120        | 4.4     | 2225                        | 315               |

Fresh Concrete Properties Analysis



## 6. CONCLUSIONS AND KEY FINDINGS

The comprehensive material characterization and mix design development phase has successfully established the baseline data for fire-resistant rubberized concrete development. Key findings include:

### **Material Characteristics:**

- Type I Portland cement shows typical composition with adequate strength development
- Natural aggregates meet all ASTM requirements for structural concrete
- Crumb rubber exhibits hydrophobic nature requiring increased admixture dosage
- Two distinct rubber size ranges provide different fresh and hardened properties

### **Mix Design Performance:**

- Workability decreases progressively with rubber content (180mm to 115mm slump)
- Air content increases from 2.1% (control) to 5.2% (20% rubber)
- Density reduction of 7.7% achieved at maximum rubber replacement
- Setting time delayed by 15-20% with rubber addition
- Superplasticizer demand increases exponentially with rubber content

### **Optimization Observations:**

- Fine rubber (1-4mm) provides better workability than coarse rubber
- Silica fume addition partially compensates for strength loss
- 10-15% rubber replacement appears optimal for balanced properties
- Rheological properties show critical threshold at 15% replacement

### **Recommendations for Phase 2:**

- Focus on 10% and 15% rubber replacement for detailed investigation
- Implement surface treatment techniques for rubber particles
- Evaluate high-temperature performance up to 800°C
- Conduct microstructural analysis of interfacial transition zone
- Develop thermal-mechanical constitutive models

## APPENDIX A: TEST STANDARDS AND METHODS

| Test                       | Standard      | Description                           |
|----------------------------|---------------|---------------------------------------|
| Cement Composition         | ASTM C114     | Chemical Analysis of Hydraulic Cement |
| Cement Fineness            | ASTM C204     | Blaine Air Permeability               |
| Aggregate Gradation        | ASTM C136     | Sieve Analysis                        |
| Aggregate Specific Gravity | ASTM C127/128 | Coarse and Fine Aggregates            |
| Slump                      | ASTM C143     | Slump of Hydraulic-Cement Concrete    |
| Air Content                | ASTM C231     | Pressure Method                       |
| Unit Weight                | ASTM C138     | Density of Fresh Concrete             |
| Setting Time               | ASTM C403     | Penetration Resistance                |
| Temperature                | ASTM C1064    | Temperature of Fresh Concrete         |
| Rubber Analysis            | ASTM D5603    | Rubber from Recycled Tires            |